

Baofeng Radio Operations for CMC-SAR

This document is specific to the Baofeng UV-5R radio. You may have a different model, in which case the keypad may be somewhat different, but the basic procedures will be the same.

Never turn on the radio without an antenna attached. Keying up (transmitting) without the load provided by the antenna will burn out the electronics of your radio.

On-off knob: Clockwise turns on. Anti-clockwise turns off. Be sure it clicks off. Note that this button has a very soft click, so occasionally you'll find that the radio has been inadvertently turned on or off by brushing against something. Keep this in mind as you decide where and how to carry the radio.

Display: Quick press of any button turns on the screen illumination. In the center, in large type, are two channel names; Channel A is above, Channel B is below. The number to the right of the channel name tells you that channel's numerical position in this list. The battery level indicator is in the upper right. To the left of the battery level indicator is a door key icon, letting you know that the keyboard is locked. *(Always keep the radio locked to avoid inadvertently changing a setting.)*



Notice the upward-pointing arrowhead to the left of the channel name (Channel A in the above photo). This is a very important item as it tells you which channel is your **primary** channel...that is, the one you will be transmitting on when you press PTT (Push-To-Talk). The other channel, your **secondary** channel (Channel B in the above photo) is in “receiving only” mode...you will hear the traffic on that channel but you will not transmit on that frequency when you key up. This accomplishes two things: you can monitor communications on two channels simultaneously and you can quickly switch your transmission between two preferred channels without searching around in the channel list.

When you are issued a radio, be absolutely certain that both displayed channels are the correct ones per your team leader AND that you've selected the correct transmit channel. If you're not sure, ask someone!

Preferred channels for CMC Ops (in order of preference):

CMCSAR1	462.5600 MHz (70cm, UHF)
CMCSAR2	462.6800 MHz (70cm, UHF)
CMCSAR3	467.7025 MHz (70cm, UHF)
VSAR16	155.1600 MHz (2m, VHF) (National Interoperability SAR frequency)

For joint CMC/HCSAR Ops:

UTAC41D	453.4525 MHz (70cm, UHF)
---------	--------------------------

Unless we are engaged in a joint operation with HCSAR and are told to use UTAC41D, do not transmit on that frequency as it is widely used by many local emergency agencies.

Keyboard Lock/Unlock: Press and hold for three seconds the # key (lower right key) to toggle between “locked” and “unlocked.” The radio should automatically lock ten seconds after the last button push, but get into the habit of locking it manually. *All of the actions explained below require you to first unlock the radio.*

To select the transmit channel: Unlock the radio. Push the blue A/B button. This moves the arrowhead between Channel A and Channel B. *Remember that when you press PTT you will only be transmitting on the channel with the black arrowhead...not on both of the displayed channels.* Once you have the arrowhead pointing to the channel on which you wish to transmit, lock the radio.

When might you need to change the transmit channel?

Scenario: You're in the field in a joint operation with HCSAR. IC has designated UTAC41D as the frequency for all ops comms, and you know that all the club radios are set to transmit on UTAC41D and to monitor CMCSAR1. You have information for one or more CMC members, but you don't want to broadcast this to the non-CMC personnel in the operation. Switch your radio to transmit on CMCSAR 1. Double check that you've done this correctly. Say something like “This is __ transmitting on Channel CMC SAR1 only”, then deliver your message. It would be wise to conclude your message by reminding them to switch their radios to transmit on CMCSAR1 if they wish to reply. When the discussion is concluded you should take responsibility to remind everyone to return their radios to transmit on UTAC41D. This is only to be used when absolutely necessary and always keep this sort of sidetalk to the absolute minimum.

Other reasons to change channels. The CMC frequencies were carefully chosen so as not to interfere with the transmissions of other users in this region. However, there may be situations where we receive transmissions from other entities or (worst case) other entities are receiving our transmissions. If you ever hear non-CMC traffic on one of our channels do not engage with the other operator, but immediately call for a Crew-wide channel switch. Another reason to switch channels may be that the terrain is such that one frequency is carrying signals better than another.

Important fact concerning radio function:

When you press PTT you “lock out” every other radio in the network (all the radios *on that channel* that receive your transmission). Also, while PTT is pressed you cannot hear any transmissions on *either of your designated channels*.

Radios set to monitor a CMC frequency as the secondary channel will not receive traffic from the primary channel if someone is already transmitting on the secondary channel.

This is why you must utilize the secondary channel very sparingly, and transmissions on either channel should be short, to the point, and limited to mission-critical information.

How to switch to a channel that is not currently displayed as either Channel A or Channel B.

IMPORTANT: In a joint ops situation, the “official” ops channel should always remain in the display and the other channel (CMCSAR1, for example) is the one you will change.

Scenario: Your Channel A is CMCSAR1 and your Channel B is UTAC41D. You’ve been asked to switch to CMCSAR2 for crew-only traffic because someone is picking up non-CMC transmissions on CMCSAR1 and we don’t want to interfere with them. Unlock the radio. Use the A/B button to put the arrowhead beside CMCSAR1. Now use the “up” or “down” arrowhead buttons in the keyboard to change the channel to CMCSAR2. You’ll see the channel names changing in the slot where CMCSAR1 used to be. Remember that our usual channels are located together in the list, so if you push the “up” button more than five or six times and you don’t see CMCSAR2, you’re going the wrong way, so start pushing the “down” button. Once you see CMCSAR2 in the display, lock the radio and you’re set to transmit on that channel.

How to change to a specified frequency rather than to a named channel.

Scenario: You’re with HCSAR on a mutual aid mission in Tennessee. The local EM unit running the operation is on TENVOL7, a channel that we don’t have programmed into our radios. IC tells you that the frequency of TENVOL7 is 457.320. You need to set your radio to receive and transmit on that frequency.

Unlock the radio. Press the orange VFO/MR button on the front of the radio. This switches the radio between what is called “channel mode” (which displays channels by name) to “frequency mode.” The display now has numerical frequencies where the channel names used to be. Everything else is the same. Use the keyboard to type in the desired frequency (4-5-7-3-2-0 in the above example). Lock the radio. The arrowhead will be beside the frequency you just entered, so that means you are now sending and receiving on 457.320 MHz along with everyone else in the operation.

1) You can set and display two different numerical frequencies by toggling with the A/B button. The radio will only remember those two frequencies (A and B); you cannot create an internal frequency list, and nothing you do in frequency mode will alter the data in your channel list, so don’t worry about “messing up” the radio’s onboard data.

2) You cannot simultaneously have one channel in frequency mode and the other in channel mode. If you need a secondary channel while in frequency mode you’ll have to use the above instructions to manually enter a secondary frequency. Remember that if other radios are not also set for this same secondary frequency, they will not receive your transmission, so this must be done before teams separate.

3) Some Baofeng radios don’t have a VFO/MR button. In that case you switch modes by turning off the radio, then hold the menu button down while you turn it back on.

4) If you follow the above instructions but your radio doesn’t work, it’s most likely due to the other team having access tones set on their frequency. You can correct this by going to menu item 13 (CTCSS) and setting the correct tone.

Other features of the radio:

Small black button, left side below PTT. Push once for flashlight, push again for flashing strobe light, push again to turn light off.

Small orange button, left side above PTT. Push once: Broadcast radio (preset to WNCW, 88.7 FM). Incoming communications from one of your two preset channels will temporarily override broadcast radio, but you must push the orange button again to prevent the broadcast from returning once the communication signal has ended. Push and hold: Alarm Mode (light flashes, siren wails). This alarm is set to local so that siren audio will not be transmitted to other radios. Could be useful, but it's not as loud as a decent mouth-operated whistle. Both of these features only work in channel mode.

Menu: Be very careful not to change any settings in the menu unless you know what you're doing. Online sources will assist with this. We do recommend that you learn how to operate the Menu by practicing adjusting the squelch, which is menu item #0 or #1. Squelch settings range from 0 to 9. Usually squelch on 5 works well. Lower numbers make the radio more sensitive to weak signals. If you are getting lots of static, squelch needs to be increased.

Band: You shouldn't have need to mess with this setting. Ban should be set to Wide for general use.

Miscellaneous:

These radios are designed to be held while in use...your body contributes to the "ground" that the radio antenna needs to receive clear signals. Generally speaking, the best signal will be achieved when holding the radio up in front of your face with the antenna pointing straight up, but terrain and other factors may be such that rotating your body, moving away from vehicles or large rocks, walking down the trail a few meters, or holding the radio at a non-vertical angle may improve signal strength. Set the unit on a table and experiment with different positions while in broadcast radio mode and you'll quickly see the effects of body contact and position on radio performance.